

Exercise – Function – PDF of the Student's t distribution

This file contains a custom MATLAB function for evaluating the PDF of the Student's t-distribution. The function takes evaluation points and degrees of freedom as inputs and returns the corresponding density values.

```
1 function y = PDF(x,nu)
2 %PDF Evaluate the PDF of the Student's t distribution.
3 % Version 1.0, September 2026
4 % Compatible with MATLAB R2014a and later
5 %
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7 %
8 % Description:
9 %   This function evaluates the probability density function (PDF) of
10 %   the Student's t distribution using its analytical expression. The
11 %   gamma function is evaluated numerically using Euler's integral.
12 %
13 % Syntax:
14 %   y = PDF(x,nu)
15 %
16 % Inputs:
17 %   x - Scalar, vector, or matrix of evaluation points.
18 %   nu - Degrees of freedom. Must be positive.
19 %
20 % Outputs:
21 %   y - Values of the Student's t-distribution PDF evaluated at x.
22 %
23 % Notes:
24 %   - The Student's t-distribution PDF is given by
25 %
26 %           Gamma((nu+1)/2)
27 %   f(x) = ----- * (1 + x.^2/nu).^(-(nu+1)/2)
28 %           sqrt(pi*nu) * Gamma(nu/2)
29 %
30 %   - The gamma function is computed using Euler's integral
31 %
32 %           Gamma(z) = integral_0^Inf t^(z-1) exp(-t) dt.
33 %
34 % ----- BEGIN FUNCTION BODY BELOW -----
35
36 % Compute Gamma((nu + 1)/2)
37 GammaNuPlusOneOverTwo = ...
38     integral(@(t) t.^((nu + 1)/2 - 1) .* exp(-t),0,Inf);
39
40 % Compute Gamma(nu/2)
41 GammaNuOverTwo = ...
42     integral(@(t) t.^(nu / 2 - 1) .* exp(-t),0,Inf);
43
44 % Evaluate the Student's t-distribution PDF
45 y = (GammaNuPlusOneOverTwo ./ ...
```

```
46     (sqrt(nu*pi) .* GammaNuOverTwo)) .* ...  
47     (1 + x.^2/nu).^(-(nu + 1)/2);  
48 end
```